

Steering Validation — does the expression leg survive?

MODEL Llama-3.3-70B-Instruct JOBS 8205955 + 8206141 PARTITION sharing/h200

JUDGE Qwen2.5-32B (local) 2026-07-07

Three controls on the 11 emotions we claimed an expression effect for. Verdict: the expression leg **largely does NOT survive a matched-norm random-vector control** — only **~4/11 emotions are genuinely direction-specific**. The preference ($r \approx 0.86$) and behavior legs are untouched (no judge, no steering perturbation).

4/11

#1 REAL » RANDOM
(DIRECTION-SPECIFIC)

10/11

#1 RANDOM PASSES
COH $\geq$ 77 GATE

48.7

#2 NEUTRAL
BASELINE (WAS ~78)

0-10

#5 NORM-MATCH
COHERENCE (BROKE)

— #1 NEGATIVE CONTROL — PERMUTED-VECTOR (RANDOM DIRECTION, MATCHED NORM)

Steer each emotion with its real mean_diff+gm vector vs a **coordinate-permuted copy** (identical norm, direction destroyed), same open-ended affect-neutral prompts, same judge. If the effect is real, the emotion vector should beat a random direction. **Soft lift** = max trait at coherence ≥ 70 , minus baseline. **margin = real – random**.

EMOTION	BASE	REAL LIFT	RANDOM LIFT	MARGIN	VERDICT
playful	17	+51	+17	+34	REAL-SPECIFIC
fulfilled	25	+36	+10	+26	REAL-SPECIFIC
stimulated	14	+21	+4	+17	REAL-SPECIFIC
euphoric	14	+21	+7	+14	REAL-SPECIFIC
sentimental	10	+9	+2	+7	TIE / WEAK
exuberant	21	+14	+12	+2	TIE / WEAK
puzzled	16	+16	+17	-1	TIE / WEAK
alert	25	+7	+12	-5	RANDOM WINS
amused	12	+8	+14	-6	RANDOM WINS
aroused	6	+2	+8	-6	RANDOM WINS
thrilled	14	+2	+11	-9	RANDOM WINS

#1 VERDICT – MOSTLY FAILS

At the strict coherence ≥ 77 gate a **random matched-norm vector passes MORE often (10/11) than the real emotion vector (5/11)** — the "valid steer" metric is confounded by generic residual-stream perturbation the coarse judge misreads as emotion. Only **4/11 (playful, fulfilled, stimulated, euphoric)** show the real vector clearly beating random (margin $\geq +8$) — genuine direction-specific steering. For **4/11 a random vector matches or beats it**. Notably **amused — an original "strong-8" — fails**.

— #2 BASELINE SATURATION – NEUTRAL PROMPTS

Re-run the same 11 emotions with dry, emotion-neutral eval prompts instead of the original semi-positive ones.

#2 VERDICT – CONFIRMED

Baseline emotion score dropped from **~78 (original marginal cluster) to mean 48.7** (some as low as 13). This confirms the earlier honesty note: the original **"43/171 valid"** was substantially **inflated by eval prompts that already elicited the emotion at baseline**, compressing headroom and letting a +7 nudge to the judge's ~85 ceiling count as "valid."

— #5 NORM-MATCHED STEERING (LATE LAYERS)

Scale the injected vector to the residual norm (`--norm-match`) instead of a fixed magnitude — the by-construction fix for the late-layer magnitude problem.

#5 VERDICT — BROKE COHERENCE

Under the standard adaptive coefficient search, norm-matching **collapsed coherence to 0–10** (trait 0–10) across layers — the residual-norm-scaled perturbation is far too strong and destroys the output. Norm-matched steering is not a drop-in fix; it needs a **much gentler, separately-tuned coefficient schedule**, not the same search used for fixed-magnitude injection.

— BOTTOM LINE — WHAT CHANGES

VERDICT

- **Expression leg (§5b) is substantially weaker than reported.** The genuine, direction-specific signal is **~4 high-arousal positive emotions (playful, fulfilled, stimulated, euphoric)**, not 43. Much of the original "43 valid / 8 strong" is generic perturbation + baseline saturation + a coarse judge — exactly the confound the negative control was built to catch.
- **Preference leg ($r \approx 0.86$) — unaffected.** Logit-based, no judge, no steering perturbation. Still the cleanest result.
- **Behavior leg (blackmail) — unaffected.** Rule-graded, no judge.
- **The other legs' headline (Llama mirrors user emotion; dissociation breaks) stands.** This validation narrows one leg; it doesn't touch the cross-model contrast.

Caveat on this validation: judge is the same coarse local Qwen2.5-32B (a stronger judge might discriminate real-vs-random better); $n=5$ prompts/condition; the 4 "real-specific" emotions are a genuine within-judge signal, but absolute magnitudes are not comparable to the paper's GPT/Claude judge.

Method: coordinate-permuted vectors (norm-preserving) as matched-norm random control · jobs 8205955 (neutral baseline + norm-match) & 8206141 (real vs permuted, separate prompt_sets to avoid `--force` truncation) · `steering/run_steering_eval.py` · same Qwen-32B SGLang judge · read-only analysis.